Model QP

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FOURTH SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION, MARCH 2026
Fourth Semester
MAJOR ELECTIVE – MG4DSEBTS200
ENZYMOLOGY
(2024 Admission Onwards)

Time: 2 Hours **Max. Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall outcome availability.

Part A

Multiple Choice Questions

Answer all questions.

Each question carries 1 mark

(10 X 1 = 10)

1. Which of the following represents an incorrect classification as per IUBMB
   (a) Oxidoreductases (b) Transferases (c) Hydrolases (d) Synthases
   [CO1, R]

2. The non protein chemical compound required for enzyme activity derived from vitamins are
   (a) Coenzymes (b) Cofactors (c) Mg^{2+} (d) Heme
   [CO1, U]

3. Holoenzyme refers to
   (a) Enzyme without prosthetic group
   (b) Prosthetic group only
   (c) Complete enzyme with prosthetic group
   (d) None of the above
   [CO1, R]

4. Which enzyme begins in the stomach continues in the small intestine and targets peptide bonds?
   (a) Amylase (b) Lipase (c) Pepsin (d) Trypsin
   [CO2, U]

5. K_m value is defined as
   (a)The maximum reaction velocity when enzyme is saturated
   (b) The substrate concentration at which velocity is half of V_{max}
   (c) The turnover number of the enzyme
   (d) The dissociation constant of the enzyme

[CO2, U]

6. In competitive inhibition, the inhibitor is
   (a) A structural analog to the substrate
   (b) A coenzyme
   (c) The prosthetic group
   (d) None of the above
   [CO2, R]

7. Which method is best suited for resolving DNA fragments by size for cloning enzyme studies?
   (a) Gel filtration chromatography
   (b) Thin layer chromatography
   (c) Agarose Gel Electrophoresis
   (d) HPLC
   [CO3, R]

8. Which enzyme is commonly measured to diagnose myocardial infarction?
   (a) Glucose oxidase
   (b) Pyruvate dehydrogenase
   (c) Streoptokinase
   (d) Creatine kinase- MB
   [CO4, R]

9. Which enzyme is industrially used for producing high fructose corn syrup/
   (a) Glucose isomerase
   (b) Lactase
   (c) Rennet
   (d) Fructose 6 Phosphatase
   [CO4, U]

10. The first step in protein purification from a cell homogenate is
    (a) Ion exchange chromatography
    (b) Centrifugation
    (c) Ammonium sulphate precipitation
    (d) Affinity chromatography
    [CO3, An]

Part B

Short Answer Type Questions

Answer any ten questions.

Each question carries 3 marks.

(10 X 3= 30)

11. Comment on the effect of substrate concentration on enzyme activity.[CO2,U]
12. What are the characteristic features of the active site in an enzyme? [CO2,R]
13. Briefly give an account on the milestones in enzymology.[CO1,R]
14. Differentiate between Fischer Model and Koshland Model of enzyme.[CO1,U]
15. What are the roles of cofactors and coenzymes in an enzyme reaction?[CO1,U]
16. Define enzyme kinetics and represent the terms in Michaelis Menten equation.[CO2,U]
17. Discuss on the differences between competitive and non competitive enzyme inhibition.[CO2,E]
18. What is feedback inhibition in enzyme catalysed reaction? Give an example.[CO2,U]
19. Write a short note on purification methods for enzyme preparation.[CO3,R]
20. Comment on the chromatographic methods employed in separating enzymes.[CO3,U]
21. List any two enzymes that are used as therapeutic agents.[CO4,R]
22. What are isoenzymes? Give an example.[CO4,U]

Part C

Short essay Type Questions

Answer any six questions.

Each question carries 5 marks.

(6 X 5 = 30)

23. Give an account on the classification of enzymes.[CO1,R]
24. What do you mean by enzyme regulation? Elaborate giving three examples.[CO2,An]
25. Write in detail on enzyme specificity with suitable examples.[CO1,U]
26. What are the principles and uses of ELISA & Western blotting techniques?[CO3,E]
27. Highlight the role of amylases, proteases and lipases in industry.[CO4,U]
28. Mention the roles of enzymes in clinical diagnosis with suitable examples.[CO4,R]
29. Explain the various strategies used for enzyme purification and extraction.[CO3,E]
30. Give a brief account on the mechanism of enzyme action.[CO2,U]